

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	M. Shanmukha Aditya	Reg. No.:	192365055
Date:		Duration:	60 minutes

Code of Conduct

I M. Shanmukha Aditya (192365055) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _M. Aditya_____

Annexure A – Agile Sprint Planning & User Story Workshop

SPRINT PLANNING DOCUMENT

Smart Automation & Cloud Architecture

SmartCloud Automation Platform (SCAP) – Agile Sprint Planning and User Story Workshop

Workshop approach. This document follows the Scrum framework. The product vision is first converted into epics and user stories, each story gets measurable acceptance criteria, the backlog is ordered by value and dependency, and the highest-priority stories are pulled into **Sprint 1** (two weeks), broken into tasks and sized with Fibonacci story points using Planning Poker.

1. Project Vision, Objectives and Stakeholders

1.1 Project Vision

Vision Element	Statement
For	Manufacturing enterprises with multiple plants that run machines and processes manually or in isolated on-premise systems.
Who need	Real-time visibility, automatic decision making and lower downtime across all plants.
The	SmartCloud Automation Platform (SCAP)
Is a	Cloud-native, secure, edge-enabled smart automation platform.
That	Collects device data automatically, monitors it live, triggers rule-based workflows and alerts, predicts equipment failures and integrates with ERP/CMMS.
Unlike	Today's paper logs, spreadsheets and plant-by-plant SCADA servers,
Our product	Gives one scalable, always-available, secure view and control of every plant, and acts on problems before they cause downtime.

Vision statement: "Every machine connected, every decision automated, every plant visible – securely, from the cloud."

1.2 Project Objectives (SMART)

- **Connect** all registered devices of the pilot plant to the cloud securely within the first three sprints.
- **Reduce** manual data logging effort by at least 30% within six months of rollout.
- **Deliver** real-time alerts within 2 seconds of a threshold violation.
- **Cut** unplanned downtime by 25–35% through predictive maintenance by the end of the pilot.
- **Achieve** 99.9% platform availability with role-based security and full audit logging.
- **Onboard** each new plant in days, not months, through a scalable cloud architecture.

1.3 Key Stakeholders and Scrum Roles

Stakeholder	Scrum Role / Type	Role and Interest in the Project
Plant Operations Head	Product Owner	Owns the product backlog, sets priorities, accepts or rejects sprint results, represents the business.
Scrum Master	Scrum Team	Facilitates ceremonies, removes impediments, coaches the team on Agile practice.
Development Team (4 developers, 1 QA engineer, 1 DevOps engineer)	Scrum Team	Design, build, test and deploy the increment; self-organise and estimate the work.
Senior Management / Sponsor	Key stakeholder	Funds the project and expects ROI, productivity and compliance outcomes.

Stakeholder	Scrum Role / Type	Role and Interest in the Project
Plant Managers and Supervisors	Key user	Monitor production and alerts; approve workflows and reports.
Machine Operators and Technicians	End user	Use dashboards and alerts daily; give usability feedback.
Maintenance Engineers	End user	Receive predictive alerts and work orders; improve repair planning.
System Administrator / IT Team	Key user	Register devices, manage users, roles and platform configuration.
Security and Compliance Officer	Stakeholder	Defines access, encryption, audit and regulatory requirements.
Data Scientists / Analysts	Stakeholder	Build predictive models and KPIs from collected data.
Cloud and Device Vendors	External	Provide cloud services, devices and support under SLAs.
Customers and Suppliers	External	Benefit from reliable delivery and better visibility of orders and parts.

2. Epics and User Stories

2.1 Epics

Epic	Name	Description
E1	Device Onboarding & Edge Connectivity	Register devices and connect plant equipment securely to the cloud through edge gateways, including offline operation.
E2	Data Ingestion & Storage	Receive, validate and store high-volume sensor data and make history available.
E3	Monitoring Dashboard & Alerts	Show live status and trends; notify users when thresholds are crossed.
E4	Rules & Workflow Automation	Let users define no-code rules, automatic work orders and approval chains.
E5	Security & User Management	Authentication, role-based access, multi-factor authentication and audit trail.
E6	Predictive Maintenance	Machine health scoring and failure prediction using ML.
E7	Energy Optimisation	Track energy use and schedule loads automatically.
E8	Integration & Reporting	Synchronise with ERP/CMMS and produce scheduled reports.
E9	Platform Operations (DevOps)	CI/CD pipeline, monitoring and infrastructure as code (enabler epic).

2.2 User Stories

All stories use the format "**As a <user>, I want <feature>, so that <benefit>**" and follow the INVEST criteria (Independent, Negotiable, Valuable, Estimable, Small, Testable).

ID	Epic	User Story
US01	E1	As a system administrator, I want to register a new device with its type, location and plant, so that it can be tracked and managed from the cloud.
US02	E1	As a maintenance engineer, I want the edge gateway to connect to the cloud over an encrypted, authenticated channel, so that machine data is transmitted securely without manual effort.

ID	Epic	User Story
US03	E1	As a plant manager, I want the gateway to buffer data and run local rules during an internet outage, so that no data is lost and critical automation continues offline.
US04	E2	As a data analyst, I want sensor readings to be ingested continuously, validated and stored, so that I always have complete and reliable data to analyse.
US05	E2	As a plant manager, I want to retrieve historical data by machine and date range, so that I can investigate past incidents and compare performance.
US06	E3	As a machine operator, I want a live dashboard showing the status of each machine, so that I can see problems instantly without walking the floor.
US07	E3	As a maintenance engineer, I want to receive an SMS / e-mail / push alert when a reading crosses its threshold, so that I can respond before a breakdown happens.
US08	E3	As a plant manager, I want trend charts of key parameters over time, so that I can spot gradual deterioration and inefficiencies.
US09	E4	As a plant manager, I want to create automation rules (if condition then action) without writing code, so that routine decisions happen automatically.
US10	E4	As a maintenance engineer, I want a work order to be created automatically when an alert occurs, so that repairs are scheduled without paperwork or delay.
US11	E4	As a supervisor, I want to approve or reject critical actions through a digital approval workflow, so that there is accountability and no e-mail chasing.
US12	E5	As a system administrator, I want to log in securely and assign roles (Admin, Manager, Engineer, Operator, Viewer), so that each user can access only what they are permitted to.
US13	E5	As a security officer, I want multi-factor authentication for privileged users, so that unauthorised access risk is reduced.
US14	E5	As a compliance officer, I want an audit log of every user and system action, so that I can prove compliance during audits.
US15	E6	As a maintenance engineer, I want a health score for each critical machine, so that I can prioritise which machine needs attention.
US16	E6	As a maintenance engineer, I want to be warned when the model predicts a likely failure, so that I can repair it during planned downtime.
US17	E7	As a plant manager, I want to view energy consumption per machine and plant, so that I can identify and reduce energy waste.
US18	E7	As a plant manager, I want heavy loads to be scheduled automatically in low-tariff hours, so that electricity cost is reduced.
US19	E8	As a operations head, I want SCAP to sync automatically with the ERP and CMMS, so that data is consistent and not entered twice.
US20	E8	As a senior manager, I want scheduled PDF/Excel reports delivered by e-mail, so that I get performance summaries without asking for them.
US21	E9	As a DevOps engineer, I want an automated CI/CD pipeline that builds, tests and deploys to staging, so that features are delivered quickly and with quality.

3. Acceptance Criteria (Given–When–Then)

Acceptance criteria are written for the six stories selected for Sprint 1 (see Section 5). Each criterion is specific, measurable and testable.

US12 – Secure Login with Role-Based Access

Scenario	Given	When	Then
Valid login	a registered user with an active account	the user enters the correct username and password	the user is logged in within 2 seconds and sees only the menus allowed for their role
Invalid login	a registered user	the user enters a wrong password 5 times	the account is locked for 15 minutes and an error message is shown without revealing which field was wrong
Role restriction	a user with the Operator role	the user tries to open the Admin console URL	access is denied with HTTP 403 and the attempt is recorded
Password security	a new user account	the password is saved	it is stored hashed (bcrypt) and never displayed in plain text

US01 – Register a New Device

Scenario	Given	When	Then
Successful registration	an administrator is logged in	they submit device name, type, plant and unique serial number	the device is saved with a unique ID and appears in the device list within 3 seconds
Duplicate device	a device with serial number X already exists	the admin registers another device with serial number X	the system rejects it with the message "Device already registered"
Mandatory fields	the registration form is open	the admin leaves a mandatory field empty	the form highlights the field and does not submit
Credential issue	a device is registered	registration completes	a unique device certificate or token is generated and shown once for download

US02 – Secure Gateway-to-Cloud Connection

Scenario	Given	When	Then
Authenticated connection	a registered gateway holds a valid certificate	it connects to the MQTT broker over TLS 1.2+	the connection is accepted and shown as "Online" in the device list within 5 seconds
Unauthorised gateway	a gateway has an invalid or revoked certificate	it tries to connect	the broker rejects the connection and logs a security event
Reconnect	an established connection drops	the network is restored	the gateway reconnects automatically within 30 seconds
Encryption	data is sent by the gateway	a network capture is inspected	no payload is readable in plain text

US04 – Continuous Data Ingestion and Storage

Scenario	Given	When	Then
Valid reading	an online device sends a reading in the agreed JSON schema	the ingestion service receives it	the reading is stored in the time-series database within 2 seconds with device ID and timestamp
Invalid reading	a message does not match the schema	the service validates it	it is rejected, moved to the dead-letter queue and an error counter is increased

Scenario	Given	When	Then
Load	a simulator sends 5,000 messages per second	the test runs for 10 minutes	no message is lost and average processing latency is under 2 seconds
Duplicates	the same message ID is received twice	the service processes it	only one record is stored

US06 – Live Machine Dashboard

Scenario	Given	When	Then
View status	devices are sending data	a logged-in user opens the dashboard	each machine card shows current status (Running / Idle / Fault) and latest values, refreshed at least every 5 seconds
Offline device	a device has sent nothing for 60 seconds	the dashboard refreshes	the machine is marked "Offline" in grey with the last-seen time
Filter	multiple plants and lines exist	the user selects a plant	only that plant's machines are displayed in under 2 seconds
Responsiveness	the dashboard is opened on a mobile phone	the page loads	the layout adapts without horizontal scrolling

US21 – Automated CI/CD Pipeline

Scenario	Given	When	Then
Build on commit	a developer pushes code to the main branch	the pipeline is triggered	the code is built and all unit tests run automatically in under 10 minutes
Failing tests	a unit test fails	the pipeline runs	the build is marked failed, deployment is blocked and the developer is notified
Deploy to staging	build and tests pass	the pipeline continues	the container image is deployed to the staging environment automatically and the health check returns 200 OK
Security scan	a dependency has a critical vulnerability	the scan stage runs	the pipeline fails and reports the vulnerable package

Definition of Ready and Definition of Done

Definition of Ready (story can enter a sprint)	Definition of Done (story is complete)
Story follows the As-a / I-want / So-that format Acceptance criteria are written and agreed Dependencies are identified and available Story is estimated by the team and small enough for one sprint UI mock-up or API contract available where needed	Code is complete, peer reviewed and merged Unit test coverage is at least 80% and all tests pass All acceptance criteria are verified by QA No critical or high security vulnerabilities in scan Deployed to staging and demonstrated to the Product Owner Documentation and API specification updated

4. Product Backlog Prioritization

4.1 Prioritization Method

The backlog is ordered using **MoSCoW** (Must, Should, Could, Won't for now) combined with a **business value score (1–10)** from the Product Owner, **dependencies** between stories, and **risk reduction** (risky integration and security work is done early). Stories with a high value-to-effort ratio and those that unblock other stories are ranked first.

4.2 Prioritized Product Backlog

Rank	ID	User Story	Value	MoSCoW	Depends on	SP	Planned
1	US21	Automated CI/CD pipeline	7	Must	None (enabler)	3	Sprint 1
2	US12	Secure login with role-based access	10	Must	US21	5	Sprint 1
3	US01	Register a new device	9	Must	US12	3	Sprint 1
4	US02	Secure gateway-to-cloud connection	10	Must	US01	5	Sprint 1
5	US04	Continuous data ingestion and storage	10	Must	US02	8	Sprint 1
6	US06	Live machine dashboard	9	Must	US04	5	Sprint 1
7	US07	Threshold alert notifications	9	Must	US04	5	Sprint 2
8	US05	Historical data retrieval	7	Should	US04	5	Sprint 2
9	US09	No-code automation rules	9	Must	US04, US07	8	Sprint 2
10	US13	Multi-factor authentication	6	Should	US12	3	Sprint 2
11	US14	Audit log	7	Should	US12	3	Sprint 3
12	US10	Automatic work-order creation	8	Must	US07, US09	5	Sprint 3
13	US08	Trend charts	6	Should	US05	3	Sprint 3
14	US03	Offline buffering and local rules	8	Should	US02	5	Sprint 3
15	US11	Approval workflow	6	Should	US09	5	Sprint 4
16	US19	ERP / CMMS synchronisation	8	Should	US10	8	Sprint 4
17	US15	Machine health score	8	Should	US05	13	Sprint 5
18	US16	Failure prediction alert	9	Should	US15	8	Sprint 6
19	US17	Energy consumption view	5	Could	US05	5	Sprint 6
20	US20	Scheduled reports	5	Could	US05	3	Sprint 7
21	US18	Automatic load scheduling	4	Could	US17, US09	8	Later

Total backlog size: 116 story points across 21 stories. At a forecast velocity of about 30 points per sprint, the full backlog is expected to take 6–7 sprints; the Product Owner may re-order it at any sprint review.

Rationale for the order

- **Foundation first:** CI/CD (US21), login (US12) and device registration (US01) are prerequisites for every other story.
- **Highest risk early:** the secure gateway connection (US02) and ingestion pipeline (US04) carry the highest technical risk and unblock dashboards, alerts and analytics.

- **Visible value early:** the live dashboard (US06) gives users and sponsors a working demo at the end of Sprint 1.
- **Advanced features last:** predictive maintenance, energy optimisation and load scheduling need historical data first, so they are planned after the data foundation is stable.

5. Sprint Backlog Planning (Sprint 1)

5.1 Sprint Details and Capacity

Item	Detail
Sprint length	2 weeks (10 working days)
Team size	6 members: 4 developers, 1 QA engineer, 1 DevOps engineer (Product Owner and Scrum Master are separate)
Available hours	6 members × 10 days × 6 productive hours = 360 hours
Ceremony and overhead time	Approximately 36 hours (planning, daily stand-ups, refinement, review, retrospective)
Net capacity	324 hours
Forecast velocity	About 30 story points (assumed from team capacity and a focus factor of 0.7 for a new team)
Committed scope	29 story points (6 stories)

5.2 Selected Stories for Sprint 1

ID	Story	SP	Why selected
US21	Automated CI/CD pipeline	3	Enables continuous integration for all later work
US12	Secure login with role-based access	5	Security foundation; needed by all screens
US01	Register a new device	3	Needed before any device can connect
US02	Secure gateway-to-cloud connection	5	Highest-risk integration; validates the architecture early
US04	Continuous data ingestion and storage	8	Core data pipeline; unblocks dashboard, alerts and analytics
US06	Live machine dashboard	5	Delivers visible, demonstrable value
Total		29	Within the forecast velocity of 30 SP

5.3 Task Breakdown

Story	Task	Owner	Estimate (hours)
US21	Create Git repository, branching strategy and code-review rules	DevOps	3
	Configure CI pipeline: build and run unit tests	DevOps	8
	Containerise services and write infrastructure-as-code script for staging deployment	DevOps	10
	Add dependency and security scan quality gate	DevOps	4
US12	Design user and role database schema	Developer 1	6

Story	Task	Owner	Estimate (hours)
	Build authentication API with JWT and password hashing	Developer 1	12
	Implement role-based access middleware and account logout	Developer 1	10
	Build login and role-assignment screens	Developer 2	8
	Write unit and API tests	QA	8
US01	Define device data model and registration API	Developer 2	8
	Build device registration form and device list screen	Developer 2	8
	Generate device certificate / token on registration	Developer 3	8
	Write test cases and execute tests	QA	6
US02	Set up MQTT broker with TLS and certificate authentication	DevOps	12
	Build gateway client / simulator with auto-reconnect	Developer 3	14
	Implement certificate validation and revocation check	Developer 3	10
	Integration tests including encryption check	QA	8
US04	Design message schema and streaming topics	Developer 4	6
	Build ingestion service to consume and validate messages	Developer 4	16
	Implement dead-letter queue and duplicate handling	Developer 4	10
	Set up time-series database and retention policy	DevOps	10
	Write persistence layer to store readings	Developer 3	10
	Load test at 5,000 messages per second	QA	10
	Unit tests	Developer 4	8
US06	Create dashboard wireframes and get Product Owner feedback	Developer 2	6
	Build dashboard layout and machine status cards (React)	Developer 2	14
	Build live-data API and WebSocket updates	Developer 1	14
	Add charts and plant filter	Developer 2	10
	UI and responsiveness testing	QA	6

Total planned task effort: 263 hours against a net capacity of 324 hours (81% utilisation). The remaining buffer covers defects, code reviews, spikes and unexpected work, which keeps the sprint realistic.

6. Effort Estimation Using Story Points

6.1 Technique

The team used Planning Poker with the modified Fibonacci sequence (1, 2, 3, 5, 8, 13, 21). Story points express relative size, combining effort, complexity, risk and uncertainty, rather than hours. The simplest story, US01 (Register a device – a standard form plus API), was chosen as the reference story and given 3 points; all other stories were compared with it.

Process: the Product Owner explains the story → the team asks questions → each member privately selects a card → cards are revealed together → members with the highest and lowest votes explain their reasoning → the team re-votes until estimates converge.

6.2 Planning Poker Results and Justification

ID	Story	Votes (Round 1 → Final)	SP	Justification
US21	CI/CD pipeline	3, 3, 5, 3, 3, 2 → 3	3	Well-known tooling and templates; low uncertainty; mostly configuration work by one engineer.
US12	Login with RBAC	5, 5, 8, 3, 5, 5 → 5	5	Standard pattern but security-sensitive: password hashing, lockout, JWT and role checks need careful testing.
US01	Register device	3, 2, 3, 3, 3, 3 → 3	3	Reference story. Simple CRUD, plus certificate generation as the only extra complexity.
US02	Secure gateway connection	5, 8, 8, 5, 13, 8 → 5	5	Initial 8–13 votes reflected certificate and TLS uncertainty; after agreeing to use a managed MQTT broker and a gateway simulator, the team settled on 5.
US04	Data ingestion and storage	8, 13, 8, 8, 13, 8 → 8	8	Largest story: streaming, validation, dead-letter handling, duplicates, database and performance testing; high technical risk but well understood after discussion. Not split because the parts are not independently valuable.
US06	Live dashboard	5, 5, 8, 5, 3, 5 → 5	5	Medium complexity UI with live updates and filtering; depends on stable data API but uses familiar technology.

Sanity check: the largest sprint story (8 SP) is below the "must split" size of 13. Story US15 (13 SP, machine health score) in the backlog is flagged for splitting into smaller stories (data preparation, model training, display) during refinement before it enters a sprint.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

"Deliver a secure, working end-to-end data path: authorised users can register a device, the device connects securely to the cloud, its data is stored reliably, and the live status is visible on a dashboard – all deployed automatically to staging."

7.2 Expected Deliverables at the End of Sprint 1

Deliverable	Description
Working software increment	A deployed, tested build in staging containing login with roles, device registration, secure device connection, data ingestion and live dashboard.
CI/CD pipeline	Automated build, test, security scan and deployment to staging.
Secure messaging infrastructure	MQTT broker with TLS and certificate-based device authentication.
Data store	Time-series database storing validated readings with retention policy.
Test evidence	Unit, integration and load test reports (5,000 messages per second) with at least 80% coverage.
Documentation	API specification, architecture diagram, deployment notes and updated backlog.
Sprint review demo	Live demo using a simulated gateway sending data that appears on the dashboard.

7.3 Features Completed at Sprint End

- Users can log in and are restricted by role (US12).
- Administrators can register devices and receive credentials (US01).
- Gateways connect over encrypted, authenticated channels and auto-reconnect (US02).
- Sensor data is validated and stored with no loss at 5,000 messages per second (US04).
- Operators see live machine status on web and mobile (US06).
- Every commit is built, tested and deployed to staging automatically (US21).

7.4 Sprint Burndown (Planned)

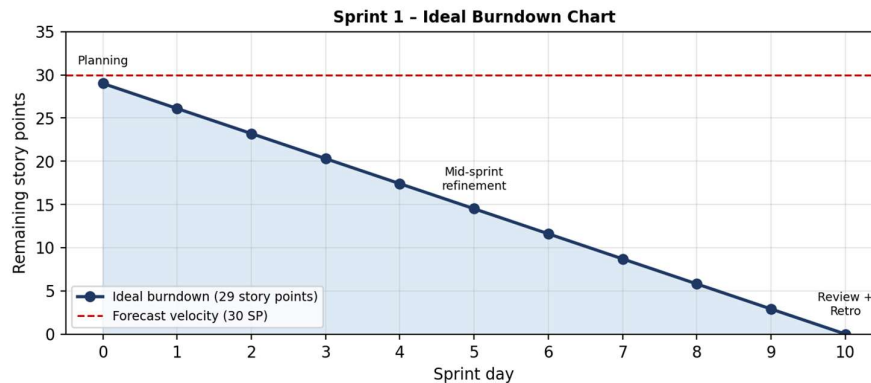


Figure 1: Planned (ideal) burndown for Sprint 1 – 29 story points over 10 days

7.5 Sprint Schedule and Ceremonies

Ceremony	When	Purpose
Sprint Planning	Day 1 (up to 4 hours)	Agree the sprint goal, select stories, create tasks and confirm capacity.
Daily Scrum	Every day, 15 minutes	What was done, what is next, any impediments.
Backlog Refinement	Day 6 (1 hour)	Clarify and estimate upcoming stories such as US07, US09, US05.
Sprint Review	Day 10 (1 hour)	Demo the increment to stakeholders and gather feedback.
Sprint Retrospective	Day 10 (45 minutes)	Identify improvements for the next sprint.

7.6 Risks and Assumptions for the Sprint

Risk / Assumption	Mitigation
Certificate and TLS set-up takes longer than planned (US02)	DevOps starts on Day 1; use a managed MQTT service as fallback.
Ingestion performance below target (US04)	Early load test on Day 6; tune batching and database indexing.
Team member unavailable	Pair programming and cross-training; the 12% capacity buffer absorbs short absences.
Requirements change mid-sprint	Changes go to the backlog; the sprint goal is protected unless the Product Owner cancels the sprint.
Assumption: simulated gateway is acceptable for the demo	Confirmed with the Product Owner at planning; real device tests start in Sprint 2.

8. Conclusion

This workshop converted the SmartCloud Automation Platform vision into a prioritised backlog of nine epics and twenty-one user stories. Six high-value, dependency-ordered stories totalling 29 story points were selected for Sprint 1, supported by measurable Given–When–Then acceptance criteria, a task plan of about 260 hours that fits team capacity, and Planning Poker estimates with clear justification. The clear sprint goal, the shared Definition of Done and the planned review give stakeholders a working, secure end-to-end data path in two weeks, while later sprints build automation, predictive maintenance and integration on that foundation.